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Department of Computer Science & Engineering (Cyber Security)

CCD 415 Main Project Phase - I Zeroth Review

Bluetooth Mesh Chat Application

Guided By,
Ms.Farbeena N
Assistant Professor
Department of CSE

Project Members

1. Abhishek KB(LJCE23CC034)
2. Shanid Ali(JCE23CC027)
3. Akshay N(JCE23CC008)



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ABSTRACT

- Bluetooth Mesh Chat Application enables messaging between nearby smartphones without the internet, Wi-Fi, or cellular networks using Bluetooth Low Energy (BLE).
- The application uses BLE mesh networking, allowing messages and GPS location data to be forwarded through intermediate devices, thereby extending communication range.
- It is useful for emergencies, remote areas, disaster response, and places with poor network connectivity.

INTRODUCTION

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- SecureMesh is an offline messaging application designed for communication without internet or cellular networks.
- The system uses Bluetooth Low Energy (BLE) mesh networking to connect nearby devices and forward messages.
- It supports secure messaging and GPS location sharing for better communication and coordination.
- The application is useful during emergencies, disasters, remote travel, and areas with poor network coverage.
- Security features such as end-to-end encryption and encrypted local storage help protect user data and privacy.

PROBLEM STATEMENT

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- Most communication applications depend on internet or cellular networks.
- Communication becomes difficult in remote areas, disasters, and network outages.
- Direct Bluetooth communication has a limited range between devices.
- Existing systems may lack secure offline messaging and location sharing.
- Therefore, a secure system is needed to provide offline communication using BLE mesh networking.

OBJECTIVES

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- To develop an offline messaging application without internet or cellular networks.
- To use Bluetooth Low Energy (BLE) mesh networking for communication between nearby devices.
- To extend communication range by relaying messages through intermediate devices.
- To enable GPS location sharing through the mesh network.
- To support private chats, group communication, and message delivery confirmation.

EXISTING SYSTEM

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- WhatsApp: Internet-based messaging with end-to-end encryption.
- Bridgefy: Offline messaging using Bluetooth mesh networking.
- Briar: Secure peer-to-peer messaging using Bluetooth or Wi-Fi.
- FireChat: Mesh-based communication without cellular networks.

PROPOSED SYSTEM

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- Develop a BLE-based offline messaging system without internet or cellular networks.
- Use Bluetooth mesh networking to relay messages through nearby devices.
- Implement end-to-end encryption for secure communication.
- Enable GPS location sharing through the mesh network.
- Support private chats, group chats, and message delivery acknowledgements.
- Provide encrypted local storage for securely storing messages.

SCOPE OF THE PROJECT

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- Develop an offline communication system using Bluetooth Low Energy (BLE).
- Enable message forwarding through nearby devices using mesh networking.
- Support GPS location sharing between connected users.
- Apply the system in emergencies, remote areas, disaster situations, and locations with poor network connectivity.
- Support GPS location sharing between connected users.
- Implement encryption and secure local message storage.

CONCLUSION

- SecureMesh provides secure communication without requiring internet or cellular networks.
- BLE mesh networking helps extend communication by forwarding messages through nearby devices.
- Encryption and secure storage protect user messages and location information.
- GPS location sharing improves coordination during emergencies and remote activities.
- The project offers a reliable and privacy-focused solution for communication in areas with limited network connectivity.

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Thank You